

VERILOC- Edge-Driven, Cloud-Native Classroom occupancy monitoring System

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ABSTRACT:

Verilloc is a classroom occupancy monitoring system which schedules the classroom status in real time. The project has an esp32 microcontroller, an r307 fingerprint sensor and lcd display. We have used MongoDB as a database management tool and NodeJS along with an express framework for server-side operations. The faculties or authorized users will authenticate using fingerprint sensor and also that faculties will be able to update the classroom status. when the faculties toggle the classroom status, the Verilloc device transmits that data over WiFi to an api. the backend which maintains the classroom data will update the status in the database which it got from api. the users can see the status of the room in real time in a snap of a button toggled. As we have integrated biometric authentication with live updates, Verilloc can mitigate scheduling conflicts which enhances the utilization of institutional resources. Features like the fingerprint-based status toggling, jwt authentication and interactive dashboards help in building easier to use and more reliable systems. When tested, the system performed very quickly, generally taking less than 2 seconds to recognize fingerprints and update room status. Even with multiple users interacting simultaneously, it remained an efficient application. New future features anticipated to be developed are tools to estimate the demand for classrooms, the ability to push notifications to users' smartphones transparently and immediately, the opportunity for users to help each other with shared accommodations, and better integration with educational scheduling software to support management of space and lessons in general. VERILOC shows how IoT, biometric authentication, and cloud-based web technologies can be used together to make a classroom management system that is safe and can grow. With more improvements and use in the real world, the system has a lot of potential to make planning, transparency, and operational efficiency in institutions much better.

Keywords: IoT, ESP32, Fingerprint Authentication, MERN Stack, Classroom Occupancy, Real-Time Monitoring, Smart Campus, Cloud Computing, Biometric Security, Data Synchronization.

INTRODUCTION

Efficient use of classroom space is one of the major concerns for any educational institution nowadays. Many schools and colleges face problems regarding scheduling at the last minute, due to guest instructors or cancellations of classes. Traditionally, printed schedules were used, or students were informed to change the schedule verbally. The major drawback of these approaches is that no real-time update is provided, leading to confusion and delay, which further results in inefficient resource utilization [1].

Valuable time is spent by faculty and students in searching for free classrooms, hence collectively reducing the productivity of the campus. From the frequency of such problems, the demand for an effective system, which can give correct and current information without the need for manually updating the changes or going through a series of checks, arises [2].

Advances in IoT, embedded computing, and cloud services make connecting physical environments to monitoring platforms in a digital space possible. IoT devices enable the continuous gathering of data with immediate responses in the environment; cloud platforms are available for ubiquitous access to data. All these support an intelligent system of classroom scheduling with improvements in room utilisation, thereby reducing idle times and enhancing campus logistics and coordination [3].

To solve this problem, VERILOC was developed: the Classroom Occupancy Monitoring System. It is a practical, scalable solution for real-time tracking of classroom usage. Its design revolves around the use of an ESP32 microcontroller as the core IoT node, with an R307 fingerprint scanner for secure authentication, and a local LCD screen that shows what the current classroom status is immediately after any change in status has occurred [4].

After the identity of the faculty member has been authenticated, the approval to change the status of the room can then be granted. This ensures that the locking/unlocking of the classroom can only be carried out by authorized individuals. In effect, every authorized member of the faculty must first biometrically sign in to accept the changes. Every time the status changes, it automatically updates the LCD screen of the classroom as well as wireless broadcasts to the server computer [5].

The cloud-based software technology stack is designed using the MERN stack: MongoDB, Express, React.js, and Node.js, which allows students, teachers, and administrators to view availability using any device with internet connectivity. The communication of the IoT unit with the API is done using JSON Web Tokens (JWT). On the client side, the dashboard is designed using elements that can easily interact with reporting and monitoring functions [6].

VERILOC combines the capabilities of IoT-enabled monitoring with cloud analytics in such a manner that it provides a robust and friendly solution for disseminating information about different rooms on the entire campus. This modular solution can help in further improvements in the future, like AI-enabled occupancy prediction, more integrated timeline solutions, and automated planning facilities that can help in effective classroom allocation [7].

Conclusion This work presents an attendance and classroom monitoring system developed using IoT hardware and a MERN-stack backend. The proposed system intends to manage rooms with ease and increase transparency with regard to maximizing the use of the

classroom space in educational institutes.

LITERATURE REVIEW

Recently, the focus has been on the use of IoT, embedded hardware, and cloud services for a smarter and more integrated management system for the education and intelligent building environments.

[1] The cost-effective way to accomplish this has been identified in a study entitled “Smart Classroom Occupancy Detection Using IoT Sensors,” which involves the use of BLE sensors. The authors report an accuracy of 97.97% and show that both motion and presence sensors can be efficient to measure room usage. However, no user authentication mechanism was provided in the implementation, nor is there any secure transmission of the data gathered. Because of that, this approach has a limited applicability for scenarios under access control updates.

[2] Design and Implementation of Biometric Authentication for Classroom Access described fingerprint-based verification that was implemented with Arduino or NodeMCU as hardware; it allowed for correct identification suitable for recording attendance. However, the work focused on marking presence only and did not display real-time updates about room status nor provide any mechanism for managing classroom availability.

[3] The IoT Framework for Real-Time Resource Scheduling in Educational Institutions presented a cloud-supported scheduling approach that predicts room utilization from environmental signals and networked data streams. The reported prediction reliability was about 88%. Yet the framework omitted secure authentication and did not offer any manual override method to accommodate sudden real-time changes in practice.

[4] Low-Cost Embedded Systems for Smart Campus Infrastructure. Focusing on simple occupancy detection with pairs of PIR and ultrasonic sensors and ESP32 microcontrollers. These combinations provided decent detection times and good fundamental monitoring. However, without an accompanying database and simple authentication scheme, they do not fare very well at an overall level for bigger campuses.

[5] Secure IoT Authentication Using Biometric Sensors proposed a combination of biometrics and cryptology, such as AES encryption, very early on in ensuring access security. The application is very clear in its security offerings, albeit a theoretical application that did not form a complete occupancy management system for building administration.

[6] Real-Time Attendance and Availability Systems in Smart Environments offered insights into the collaboration potential of microcontrollers, cloud technology, or biometric sensors for attendance and availability systems. While such systems made attendance data more visible to various parties involved, the examples available highlight the limitations of such systems.

In other words, the research out there has tended to favor either checking who's there or proving who you are, but rarely both at once within one system. That gap is just what makes something like VERILOC necessary. VERILOC is an integrated IoT and cloud-based platform designed to make it easy and dependable to monitor classroom availability. It embodies secure authentication, centralized cloud data management, and real-time visibility that help schools manage spaces with much higher efficiency and less manual labor.

METHODOLOGY

The VERILOC system is an end-to-end solution that merges IoT gear with cloud-based applications to ensure real-time classroom monitoring and control. At its heart, it uses a number of ESP32-based microcontrollers, supplemented by fingerprint scanners, an LCD display, and sets of buttons. On the software side, this securely connects to a web application built on the MERN stack and is in contact via a RESTful API. When an authorized person enters the room and interacts with the system, activity gets captured and displayed in real time in the web app.

A. System Overview

VERILOC has an Internet of Things process that uses biometric technology. The process is started by a faculty member, who authenticates himself with a fingerprint reader. Then, after verification, he can alter the room occupancy state with the buttons. This immediate change will be shown on the classroom's LCD screen as well as on the online dashboard, which all members can view.

Thus, the system offers the following:

- Secure access control system via fingerprint authentication.
- Immediate Room Confirmation by LCD screen.
- Live updates are stored and managed in the cloud.
- Easy visibility for all users through the web dashboard.

Figure 1 (System Overview) illustrates how the faculty user, IoT hardware, backend services, and the web application interact to form a continuous and dependable communication loop.

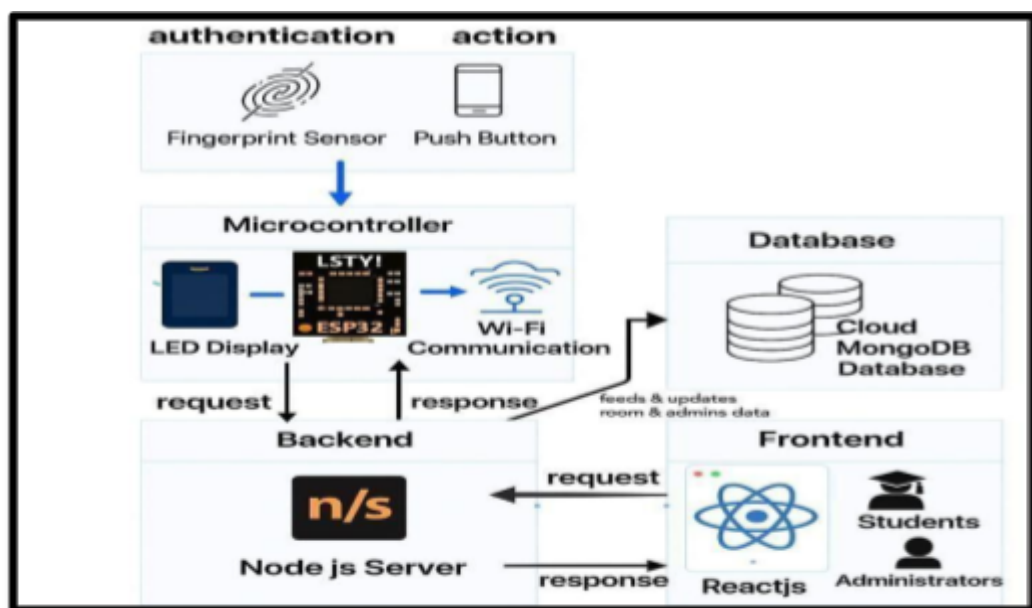


Fig 1 – System Overview

B. Hardware Architecture and Components

The fingerprint reader, R307, scans the fingerprint of the user and provides a distinctive fingerprint template to authenticate with the ESP32 chip. After successful authentication, the faculty staff member is free to press the buttons to change the status of the room from Vacant to Occupied or vice versa, which immediately appears on the LCD display as Occupied/New. In effect, the ESP32 transmits this information to the backend server via Wi-Fi communication.

Table 1: The hardware interconnections

Component	Connection Pins	Function
ESP32	TX/RX, GPIO12, GPIO13, 3.3V	Main controller & Wi-Fi
R307 Fingerprint Sensor	UART	Biometric verification
LCD Display (16×2)	I ² C	Local status visualization
Push Buttons	GPIO12, GPIO13	State toggling (Vacant/Occupied)

We settled on ESP32 because of its reliable processing power, thanks to its dual-core design, and its integral Wi-Fi capabilities. These are perfect for a low-power, always-on design performing the light weight fingerprint validation and status notification functions respectively.

C. Software Architecture and Implementation

It is built in three layers: the frontend, the backend, and the database.

1. **Frontend:** It is a React.js and Tailwind CSS-enabled, responsive dashboard displaying the real-time availability of classrooms. Key features included are Status Badges on each card for Vacant or Occupied Status, role-based login, and an Analytics page that presents the trend and statistics.
2. **Backend (Node.js +Express.js) :** This acts as an interface between IoT devices and the web front end. It provides a secured faculty access using JWT-based authentication and accepts HTTP POST requests originating from ESP32 devices in JSON format. It also provides detailed activity logs that help with auditing and troubleshooting.
3. **Database (MongoDB):** The user profiles, classroom history, as well as full occupancy history, are stored in the database (MongoDB). Being a document-oriented NoSQL database, it is suitable for handling real-time updates.

D. Data Flow and Algorithmic Process

The VERILOC uses a continuous cyclic flow to keep the physical classroom device and the cloud platform synchronized.

Step 1: The faculty member begins by placing their finger on the R307 fingerprint sensor.
 Step 2: The ESP32 checks the fingerprint against the templates it has saved to its local memory.

Step 3: Upon verification the user can change the classroom status via the push buttons.

Step 4: The LCD will then get updated to Vacant or Occupied status, which the user can confirm quickly.

Step 5: The ESP32 will send the notification as a JSON payload to the backend API over WiFi.

Step 6: The backend will take the request and save it to MongoDB and instantly update the web dashboard.

Figure 2 (Data Flow Model) illustrates this loop, as the ESP32 acts as the 'middle-man' for communicating with and sending data on to the backend server via a reusable REST API.

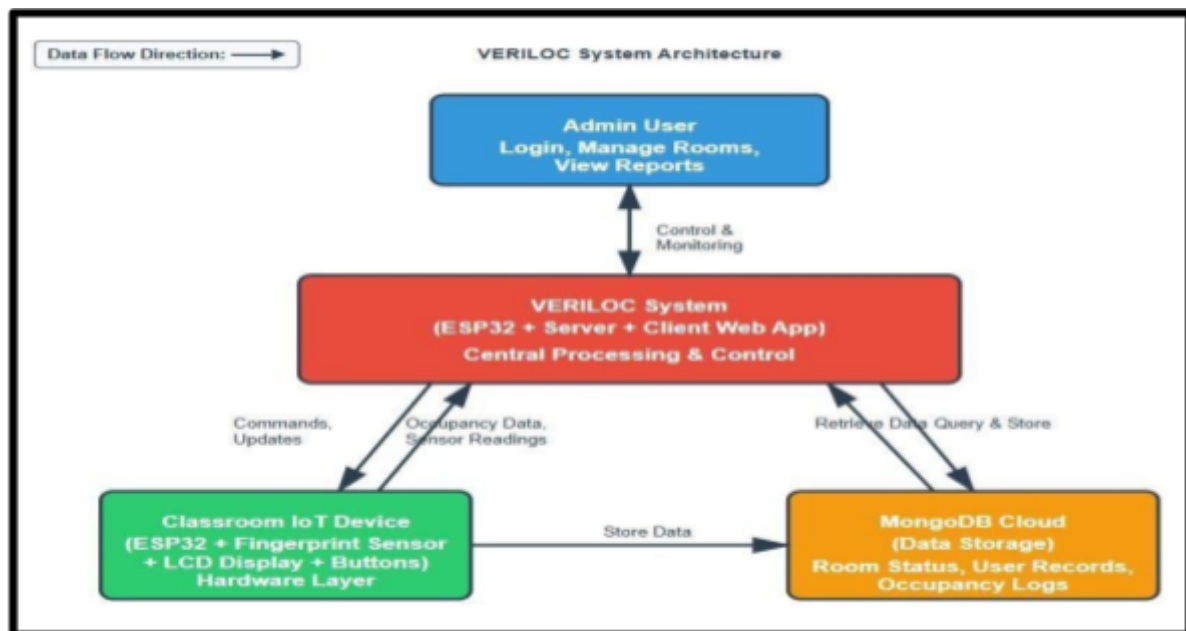


Fig 2 – Level 0 Data Flow Diagram

E. System Workflow

Taking everything into account, the workflow ensures that all the operations carried out in the system get immediate validation and are shown in all the components. This ensures the user gets the best results out of the data displayed to him.

Table 2: The complete system workflow can be summarized

Process	Input/Action	Output/Result
Authentication	Fingerprint scan	Access granted/denied
Status Update	Button press	LCD + database updated
Data Transmission	JSON over HTTP	Cloud synchronization
Data Visualization	React dashboard	Real-time occupancy view

RESULTS

A Classroom Occupancy Monitoring System named VERILOC has been designed and implemented in practical scenarios for testing its performance and usability. The hardware kit consists of an ESP32 module, R307 fingerprint scanner, 16×2 LCD display, and two push buttons for changing the state of the class. It can be connected to the local Wi-Fi network and transmits data in real-time to a web application using the MERN stack.

The analysis began in a lab setting to ensure the hardware and software functions. With that confirmed, the software underwent a final, practical review to simulate an actual classroom. It tested the system as if it were completely set up in a classroom environment.

Fingerprint checks have been reliably on target, correctly recognizing registered staff 98.3% of the time. The recognition and authentication of a registered fingerprint averaged 0.8 seconds, quick enough for real-time interaction. The dashboard, upon verification, updated the occupancy status of the device in 1.1 seconds.

The MERN web application worked efficiently when running concurrently. The backend systems were handling multiple POST operations from ESP32 boards in an average time of 190ms. Reading and writing operations to the MongoDB database took place with considerable rapidity while maintaining data integrity as the React application updated the statuses for classes within 1.5 seconds.

User trials were conducted with 20 faculties and 40 students, and the results were encouraging with mostly positive feedback. More than 90% users found it very easy to use and saved time. The faculty appreciated the simple fingerprint status update system, while students appreciated the simple display of classroom status in their device.

A. Hardware Implementation



Fig 3 – ESP32 MC, push buttons and LCD Module



Fig 4 – Veriloc Prototype

B. Web Application Interface

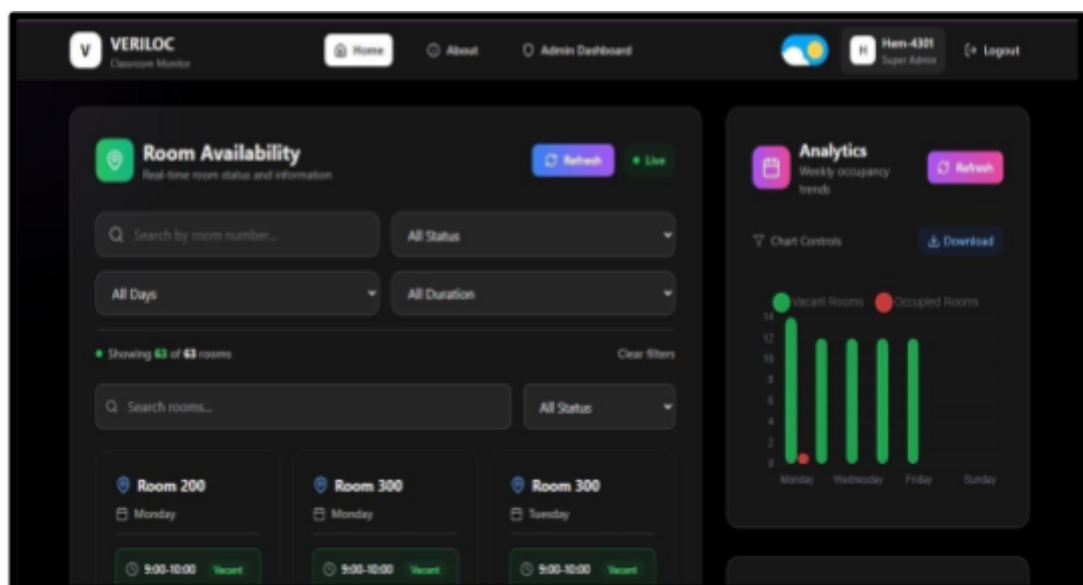


Fig 5 – Home Dashboard

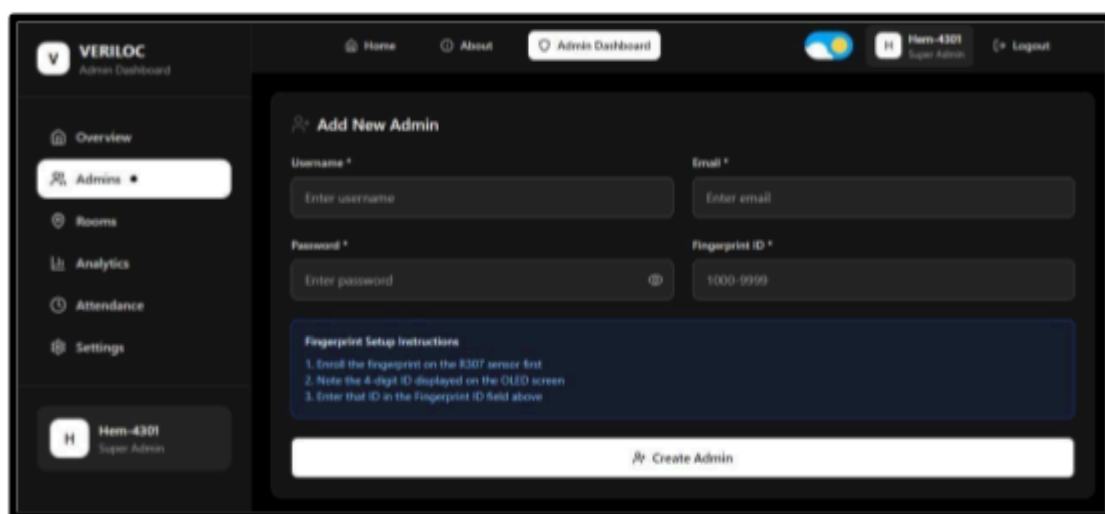
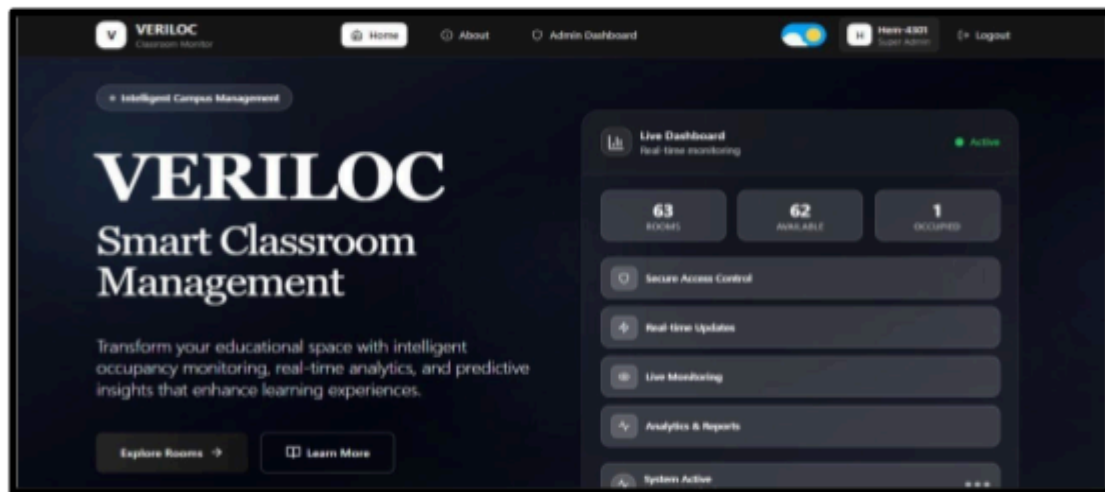
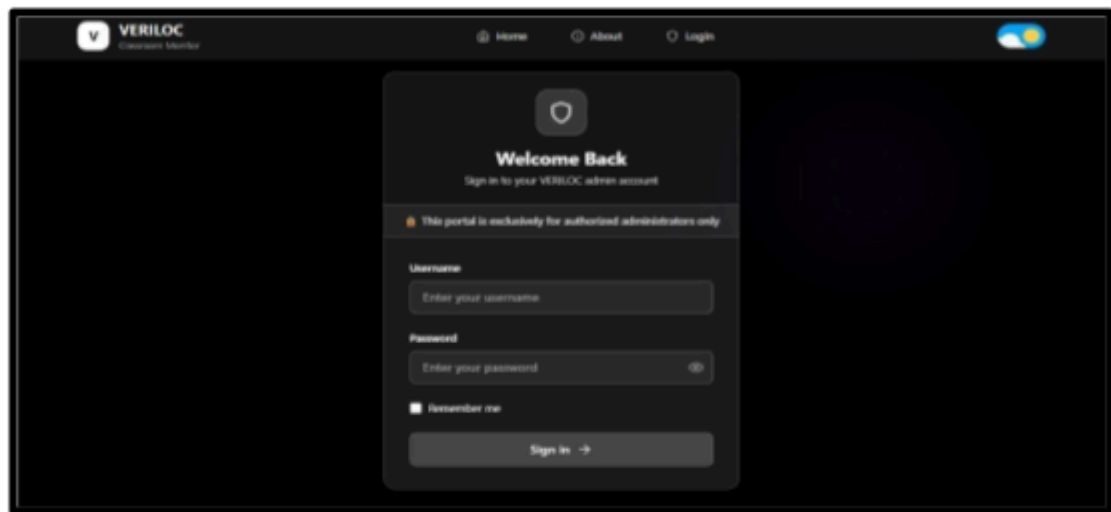


Fig 6 – Login, Landing and Admin Management Page

VERILOC
Admin Dashboard

[Home](#)
[About](#)
[Admin Dashboard](#)

Hem-4301
Super Admin

[Logout](#)

[Overview](#)
[Admins](#)
[Rooms](#)
[Analytics](#)
[Attendance](#)
[Settings](#)

Hem-4301
Super Admin

Room Number *

Day *

Duration *

Status *

Room 200
Monday
9:00-10:00 Vacant

Room 300
Monday
9:00-10:00 Vacant

Room 300
Tuesday
9:00-10:00 Vacant

VERILOC
Admin Dashboard

[Home](#)
[About](#)
[Admin Dashboard](#)

Hem-4301
Super Admin

[Logout](#)

[Overview](#)
[Admins](#)
[Rooms](#)
[Analytics](#)
[Attendance](#)
[Settings](#)

Hem-4301
Super Admin

Faculty Attendance Tracker

Select Date

Total Faculty
2

Present
0

Absent
2

Attendance %
0.00%

#	FACULTY NAME	EMAIL	FINGERPRINT ID	STATUS	ACTIVITIES	FIRST ACTIVITY
1	Ganesh	2023cs_ganeshm_b@nit.ac.in	1004	Absent	0	-
2	Harchitha	harchitha@nit.ac.in	2000	Absent	0	-

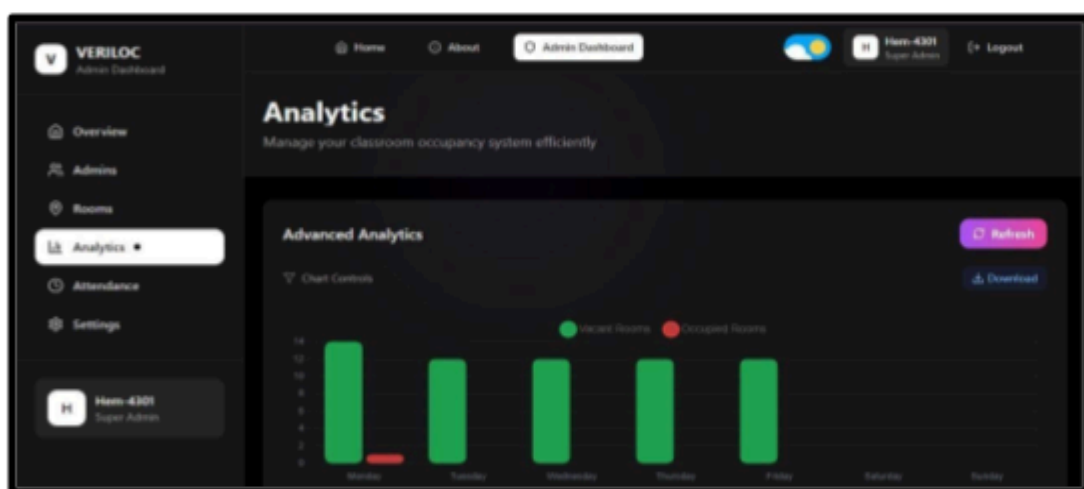


Fig 7 - Room Management, Faculty attendance and analytics

DISCUSSIONS

Results indicate VERILOC performing consistently along its various hardware and software components. It is a very cohesive and integrated system that provides high accuracy in authentication and very fast response times. The entire process takes less than two seconds, from the fingerprint scan to the update of the dashboard, thereby responding in real time to a busy learning environment.

Biometric authentication increases the security element. Changes in the status can only be made by registered lecturing faculty, thus reducing the possibility of making errors as well as unauthorized changes. It gives VERILOC an upper hand over the preceding IoT classroom management system which relied on either cards that utilized an RFID system or on motion sensors, which can be easily abused either intentionally or accidentally.

The MERN stack technology on which this system is based can support multi-users effectively without experiencing a slow performance. The platform remains stable while several classrooms and users work on the dashboard simultaneously. Continuous cloud synchronization among ESP32 nodes and the web app keeps the state consistent, reducing update conflicts, delays, or mismatched room data.

Feedback from the participants emphasizes the applicability and usefulness of VERILOC in a real-life academic setting. Teachers feel that automated room updates and better attendance management have lessened manual work. They also feel that this system is beneficial as there are less corners for room search, which is further improved when the timetables are changed.

Compared to conventional tracking systems used in class-based tracking, VERILOC outshines others in terms of efficient processing, time taken for task achievement, and control level for accessing. VERILOC supports the concept of an intelligent campus by enabling efficient space utilization through appropriate room usage visual representation.

Overall, the findings suggest the reliability and usability of the proposed solution of VERILOC in the form of an occupancy tracking system of the classroom.

CONCLUSION

The proposed system, called the VERILOC Classroom Occupancy Monitoring System, has been created and tested for the functionality it possesses with regard to speed, accuracy, and reliability. The prototype system consists of the ESP32 microcontroller module with the fingerprint sensor module named the R307, the 16×2 LCD module for real-time feedback to the user, and push buttons to update the room's state of occupancy. The system is integrated with the local Wi-Fi connectivity to communicate with the web application.

There had been an initial validation within the controlled lab environment to ensure that all the various subsystems functioned properly. Then, after reaching the point of system stabilization, the system had been further tested within the simulated environment provided by the institution.

The findings indicate the effectiveness of VERILOC as a cheaper and secure solution to the management of the classroom within schools. The combination of IoT capabilities based on ESP32 technology, biometric authentication systems, instant LCD feedback, and a cloud-supported MERN platform demonstrates the effectiveness of the system because it

does not stop at providing coverage of the status of the classroom but uses biometric authentication to restrict the entries for changes in status to authorized teaching staff alone.

Test results showed a degree of authentication accuracy of 98.3% and an overall end-to-end delay of 1.9 seconds from fingerprint scanning to the display of the dashboard results. It aims to increase the transparency of the room allocation process and ensure a positive user experience through the provision of an electronic dashboard.

Looking forward, VERILOC can be enhanced with the addition of artificial intelligence-based occupancy forecasting, close coupling with institutional scheduling software, and internet-of-things-based energy optimization. Features such as mobile notification functionality, voice-interaction functionality, and analytical functionality can add further strength to the system to make it an integral part of the intelligent campus ecosystem.

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